

The Forward Process

DATA 443/643 · Week 12, note 12.1

Week 12

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Ideas from Stanford CS231N 2025 L14 (corrupt x ; jump to x_t in one shot). Original slides; that course is not copied.

Lecture plan

- 1 A fixed kernel. No discriminator.
- 2 α_t and $\alpha\text{-bar}_t$
- 3 The one-line reparameterization
- 4 Print $\alpha\text{-bar}$ at 1, $T/2$, T

Key goal: training samples t and jumps. You do not walk $1 \dots t$.

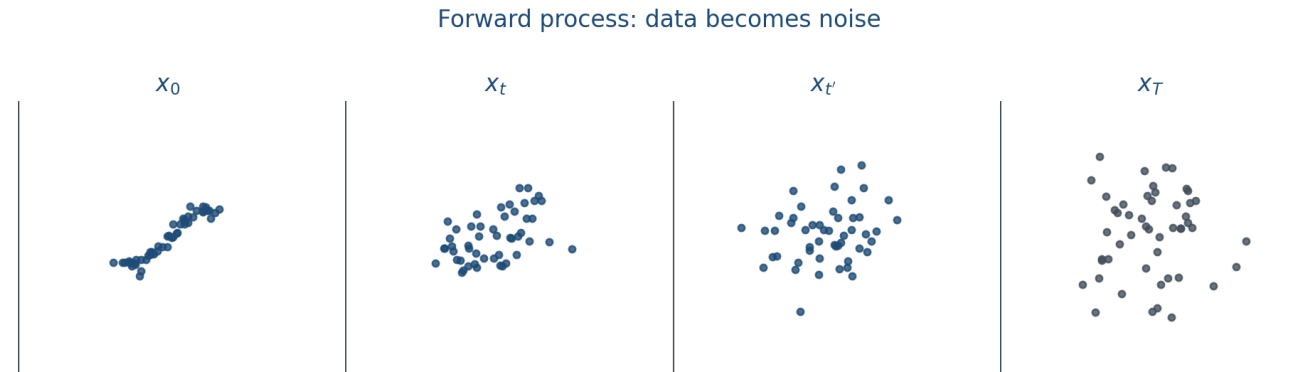
First time: what / why / formula

- What / why. Forward process $q(x_t | x_0)$ is a fixed Gaussian that interpolates from data ($t = 0$) to nearly isotropic noise ($t = T$). You can jump to any t in one shot. GANs fight a min-max. Diffusion trains a denoiser with MSE. The kernel is not learned.
- Architecture. Markov chain $x_0 \rightarrow x_1 \rightarrow \dots \rightarrow x_T$. Variance schedule β_t or $\bar{\alpha}_t \in [1, 0]$.
- Formula. $q(x_t | x_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t}x_0, (1 - \bar{\alpha}_t)I)$.
- Tradeoffs. + Stable training target, likelihood-related. – Forward is easy; reverse sampling is slow (next notes). Not a GAN.

How. Sample $\varepsilon \sim \mathcal{N}(0, I)$, set $x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\varepsilon$. Lab 12 does this in 2-D.

Destroy the sample on purpose

- $\alpha_t = 1 - \beta_t$, $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$.
- Markov:
 $q(x_t | x_{t-1}) = \mathcal{N}(\sqrt{\alpha_t} x_{t-1}, \beta_t I)$.
- The forward process is not learned.



Closed form, one Gaussian

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \varepsilon$$

$\varepsilon \sim \mathcal{N}(0, I)$.

$\bar{\alpha}_t = 1$ keeps x_0 . Zero keeps ε .

That one-liner is the whole point of this note.

A scalar path

- $x_0 = 1, \varepsilon = 2$.
- At $\bar{\alpha} = 0.64, x_t = 2.0$: more noise than signal.
- Lab 12: same square roots on a 2-D vector.

$x_0 = 1, \varepsilon = 2$. At $\bar{\alpha} = 0.64$ you already sit at 2.0.

$\bar{\alpha}_t$	$\sqrt{\bar{\alpha}} x_0$	x_t
1	1.0	1.0
0.64	0.8	2.0
0	0	2.0

Do not simulate the chain at train time

Markov loop

Walk t hops of $q(x_s | x_{s-1})$. Slow, and unnecessary for the loss.

Jump

Sample t , draw ϵ , form x_t . Noise levels, then learn to undo a bit.

What to take from this lecture

Three names

β_t , α_t , $\bar{\alpha}_t$. Write them every time.

Too fast

Every $\beta=0.9$: early t is already junk.

Lab 12

Print $\bar{\alpha}$ at 1, $T/2$, T . They should fall.